

AIME 2006 Problem 6

Problem: Let S be the set of real numbers that can be represented as repeating decimals of the form

$$0.\overline{abc},$$

where a, b, c are distinct digits. Find the sum of the elements of S .

Solution:

Each element of S has the form

$$0.\overline{abc} = \frac{100a + 10b + c}{999},$$

where a, b, c are distinct digits chosen from $0, 1, \dots, 9$.

There are

$$10 \cdot 9 \cdot 8 = 720$$

ordered triples (a, b, c) , so $|S| = 720$. We use the fact that

$$\text{sum} = (\text{number of terms})(\text{average value}).$$

Let

$$X = \frac{100a + 10b + c}{999}$$

be a randomly chosen element of S . Then

$$\sum_{x \in S} x = |S| \mathbb{E}[X].$$

It remains to compute $\mathbb{E}[X]$. Since each digit appears equally often in each position, the average value of a , b , and c is the average of the digits $0, 1, \dots, 9$. Hence

$$\mathbb{E}[a] = \mathbb{E}[b] = \mathbb{E}[c] = \frac{0 + 1 + \dots + 9}{10} = \frac{45}{10} = \frac{9}{2}.$$

By linearity of expectation,

$$\mathbb{E}[X] = \frac{100\mathbb{E}[a] + 10\mathbb{E}[b] + \mathbb{E}[c]}{999} = \frac{100 \cdot \frac{9}{2} + 10 \cdot \frac{9}{2} + \frac{9}{2}}{999}.$$

Therefore,

$$\mathbb{E}[X] = \frac{111 \cdot \frac{9}{2}}{999} = \frac{\frac{999}{2}}{999} = \frac{1}{2}.$$

Thus the desired sum is

$$\sum_{x \in S} x = 720 \cdot \frac{1}{2} = 360.$$

Therefore,

$$\boxed{360}.$$